Updating parameters

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Teach algorithm correctness separately from decreasing loss.

Make original states and all gradient entries explicit;restart comparisons and recompute predictions/losses after updates.

Keep one habit: Assess the independent gradient,new parameters and checked loss using the supplied key.

MAIN LESSON · SECTION 1

Prepare the connected mathematical investigation

> Keep explanation,shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Apply a **learning rate** to an ordered gradient

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

The residual is -1 and the gradient is [-4, -2].

Opening retrieval

Without opening the old lesson, explain one key idea from Gradients with a numerical check. Give an example and a mistake that the idea helps you avoid.

Look for: Estimate one partial derivative while other parameters stay fixed

Reference example, not the only acceptable wording: Teacher explanation: **Prediction** 3, residual 0, loss 0, gradient [0, 0].

Without opening the old lesson, explain one key idea from Loss functions. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate per-example squared error and mean squared error

Reference example, not the only acceptable wording: Teacher explanation: Residuals are -1 and 2, squares 1 and 4, MSE 2.5.

Without opening the old lesson, explain one key idea from Composing layers. Give an example and a mistake that the idea helps you avoid.

Look for: Trace how one layer supplies the next layer **input**s

Reference example, not the only acceptable wording: Teacher explanation: Scores are -2 and 2; activations are 0 and 2; y = 0 - 2 + 1 = -1.

Pictured end goal

By the end, I can explain updating parameters using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Gradients with a numerical check

Loss functions

Composing layers

For the compact alternative schedule, use two 45-minute sessions and keep old and new parameter boxes separate. This lesson checks one update, not a full optimiser.

Treat the mixed-state procedure as different from the stated algorithm, rather than claiming all sequential methods are invalid.

Outcomes

Apply a **learning rate** to an ordered gradient

Update all parameters from the same saved state

Verify the new loss and identify **overshoot**ing

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Model saved-state updates and actual loss checks

> Keep main target3,picture target5and independent target4fixtures distinct.

Main update,complete rates and guided key

Gradient descent uses new parameter = old parameter - **learning rate** times its gradient entry. The minus sign chooses the locally downhill direction for a positive **learning rate**.

A negative gradient therefore increases the corresponding parameter.

Keep the Week 10 example: x = 2, target 3, y = 2w + b and L = (y - 3) squared. At [w, b] = [1, 0], the loss is 1 and gradient is [-4, -2].

With **learning rate** 0.1, new w = 1 - 0.1 times -4 = 1.4 and new b = 0 - 0.1 times -2 = 0.2.

This particular step reaches zero loss on one example. It does not establish a unique parameter solution, successful learning on other examples or a generally ideal **learning rate**.

Predict the rows: unchanged parameters and loss 1 at rate 0; [1.2, 0.1] and loss 0.25 at rate 0.05; [1.4, 0.2] and loss 0 at rate 0.1; [2.2, 0.6] and loss 4 at rate 0.3.

The helper performs one step on this fixed example; it assumes numeric **input**s and does not choose a **learning rate** for you.

Fresh start[w, b]=[1, 0]for every row · Saved gradient · New[w, b] · New prediction 2w+b · New loss(target 3)

> **Fresh start[w, b]=[1, 0]for every row:** Rate0
>
> **Saved gradient:** [−4, −2]
>
> **New[w, b]:** [1, 0]
>
> **New prediction 2w+b:** 2
>
> **New loss(target 3):** 1

> **Fresh start[w, b]=[1, 0]for every row:** Rate.05
>
> **Saved gradient:** [−4, −2]
>
> **New[w, b]:** [1.2, .1]
>
> **New prediction 2w+b:** 2.5
>
> **New loss(target 3):** 0.25

> **Fresh start[w, b]=[1, 0]for every row:** Rate.1
>
> **Saved gradient:** [−4, −2]

> **New[w, b]:** [1.4, .2]
>
> **New prediction 2w+b:** 3
>
> **New loss(target 3):** 0

> **Fresh start[w, b]=[1, 0]for every row:** Rate.3
>
> **Saved gradient:** [−4, −2]
>
> **New[w, b]:** [2.2, .6]
>
> **New prediction 2w+b:** 5
>
> **New loss(target 3):** 4

These are four separate one-step experiments, not four consecutive updates. Each restarts from[1, 0]and computes both gradient entries before either parameter changes. Rate0is a no-change comparison; positive rates choose the negative-gradient direction, but a finite step can overshoot. The function is specific to x2, target 3 and assumes numeric inputs.

It does not validate a rate, select one automatically or train on multiple examples. Distinct parameter pairs can also fit one example:both[1.4, .2]and[1, 1]predict 3.

Guided: [1.2, 0.1] gives **prediction** 2.5, residual -0.5 and loss 0.25. Exit: no.

The mixed-state procedure lowers this loss but does not implement the specified simultaneous gradient step; compare states and gradients, not just whether loss decreased.

Complete mixed-state contrast

Main start[1, 0], rate.1	Saved simultaneous procedure	Mixed-state procedure
w gradient evaluated at old[1, 0]	−4	−4
New w	1.4	1.4
b gradient comes from	Old[1, 0]:−2	New w1.4/old b0:−.4
New b	.2	.04
New prediction /loss	3 /0	2.84 /.0256

Main start[1, 0], rate.1	Saved simultaneous procedure	Mixed-state procedure
Meaning	Matches the declared simultaneous update	Different update procedure despite lower loss

Pictured target5facts and scope

Use one explicit gradient update and then recompute the loss.

Model y=2w+b, target 5. Start w=1, b=1; **learning rate** 0.05. Initial y=3, error y–target=–2, squared loss 4.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture changes the target and start: x2, target 5, start[w, b]=[1, 1], prediction 3, residual–2, loss 4, gradient[–8, –4]. Rate.05 gives[1.4, 1.2], prediction 4, loss 1. Restarting from that original start with rate.2 gives[2.6, 1.8], prediction 7, loss 4:an equally distant overshoot. Do not use the main target 3 gradient[–4, –2]in this fixture.

Both cases compute all new values from the same old state and check the new loss.

Trace the checked rate change

AI ACADEMY / AI-5/11 · 02 · TRACE THE CASE

Follow the deciding steps

01 Compute gradient

Gradient for [w, b]=[2×(-2)×2, 2×(-2)]=[-8,-4].

02 Update simultaneously

New w=1-0.05×(-8)=1.4; new b=1.2.

03 Check new loss

New y=4, residual -1, loss 1: lower than 4.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compute gradient	Gradient for [w, b]=[2×(-2)×2, 2×(-2)]=[-8,-4].
2. Update simultaneously	New w=1-0.05×(-8)=1.4; new b=1.2.
3. Check new loss	New y=4, residual -1, loss 1: lower than 4.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Gradient descent subtracts **learning rate** times gradient. A negative gradient makes this subtraction increase the parameter. Compute both gradients at the original [w,b] before updating either one.

The small step brings the **prediction** closer to five, reducing squared error. A larger step can cross the target and undo that gain, so the new **prediction** and loss must be recalculated.

One successful toy update is not **evidence** of convergence for a real network or generalisation to unseen data.

Change: Set **learning rate** 0.2 from the same starting point.

Prediction: New w=2.6, b=1.8, y=7, squared loss 4: no improvement.

Reason: The larger update jumps to an equally distant point on the opposite side of the target.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Learning rate	A multiplier that controls the size of a gradient update
Gradient descent	Updating parameters in the direction opposite the gradient
Simultaneous update	Computing all new parameter values from the same old state
Overshoot	A step that moves too far and can increase the objective

Use one gradient for one parameter snapshot

STAGE	SAVED VALUES	ACTION
Before the step	w is 1 and b is 0	Compute gradient [-4, -2]
Apply rate 0.1	Use both old parameters	Produce [1.4, 0.2]
After the step	Use both new parameters	Recompute loss 0

Read top to bottom. All gradient entries come from the same old model. The final row evaluates the fully updated model rather than mixing old and new coordinates.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Use one gradient for one parameter snapshot

STAGE	SAVED VALUES	ACTION
Before the step	w is 1 and b is 0	Compute gradient [-4, -2]
Apply rate 0.1	Use both old parameters	Produce [1.4, 0.2]
After the step	Use both new parameters	Recompute loss 0

Read top to bottom. All gradient entries come from the same old model. The final row evaluates the fully updated model rather than mixing old and new coordinates.

Starting again from [1, 0], use learning rate 0.05. Show both new parameters, the new prediction and the new loss.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

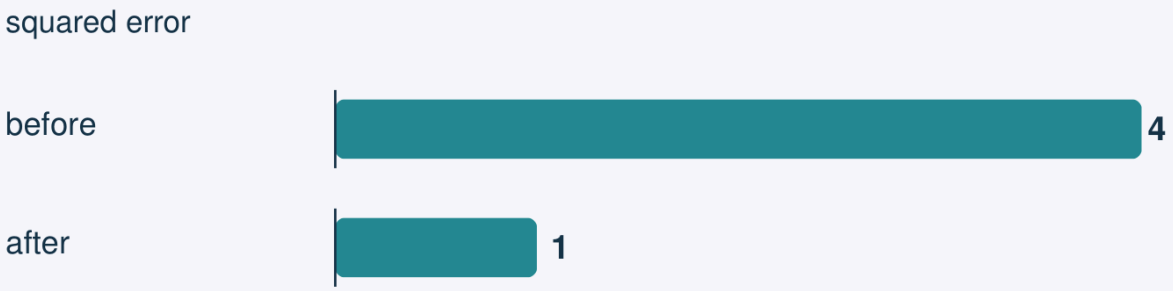
Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Updating parameters

The facts we will use

AI ACADEMY / AI-5/11

Loss before and after rate 0.05 update



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/11

Updating parameters

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute gradient

Gradient for [w, b]=[2 x (-2) x 2, 2 x (-2)]=[-8,-4].



2. Update simultaneously

New w=1-0.05 x (-8)=1.4; new b=1.2.



3. Check new loss

New y=4, residual -1, loss 1: lower than 4.

CHANGE ONE THING

Set learning rate 0.2 from the same starting point.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner's reasoning before showing a key.

Current completion task and checked reasoning

Completion task: For the lesson start [1, 0], apply rate 0.3 and calculate **prediction** and loss.

Teacher check: Teacher explanation: Gradient [-4, -2] gives [2.2, 0.6], **prediction** 5 and loss 4, larger than the starting loss 1.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Updating parameters. Proposed explanation: "Subtracting a negative gradient decreases the parameter". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Subtracting a negative adds a positive amount; show the signed calculation.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Why must new loss be recomputed rather than assumed smaller? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

A gradient supplies a local direction, but step size can **overshoot**. Rate 0.2 gives the same loss 4 despite using the correct gradient. Changed-case check: New w = 2.6, b=1.8, y = 7, loss 4: no improvement.

The larger update jumps to an equally distant point on the opposite side of the target.

Changed-case reasoning: New w=2.6, b=1.8, y=7, squared loss 4: no improvement. The larger update jumps to an equally distant point on the opposite side of the target.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Use separate old, gradient and new columns. Extension: for the lesson example, show that the new residual is the old residual multiplied by (1 - 10 times rate).

Explain why rates 0 and 0.2 preserve loss while 0.3 raises it for a nonzero residual.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve the Week 10 gradient and write the update rule with parentheses around negative entries.
7 to 18 minutes	Calculate the rate 0.1 worked step and recompute the loss from the new parameters.
18 to 32 minutes	Complete W1 and the guided rate 0.05 calculation.
32 to 45 minutes	Run or trace all four code rows and complete W2.
Session 2 start to 8 minutes	Rebuild the old-state snapshot and compare it with the mixed-state example.
8 to 20 minutes	Complete W3 and discuss zero gradient with W5.
20 to 35 minutes	Collect W4 independently, including old loss, both new parameters and new loss.
35 to 45 minutes	Use the exit response and a fresh retry. Keep the model-quality claim separate from implementation correctness.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main complete rate table and mixed-state distinction supplied. W2 rate.3 givesnew[2.2, .6], y5, L4, overshoots. W3 all gradients must use saved old state. W5 zero gradient leaves parameters unchanged, not a general global-minimum proof. Independent x1, target 4, start[1, 1]:y2, e−2, L4, gradient[−4, −4]; rate.1 new[1.4, 1.4], y2.8, e−1.2, L1.44.

Keep raw floating-point results with appropriate rounding, not literal equality to every decimal.

Diagnose the current misconceptions

Subtracting a negative gradient decreases the parameter: Subtracting a negative adds a positive amount; show the signed calculation.

A downhill direction allows any rate: A finite step can **overshoot** and increase loss.

Recompute each gradient after its assignment: For this algorithm, all entries are computed at the same saved state.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new schema or calculation transfer

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Use x = 1, target 4, w = 1, b = 1, squared loss and **learning rate** 0.1. Compute **prediction**, residual, gradient [2 times residual times x, 2 times residual], new parameters and new loss.

Teacher explanation: **Prediction** 2, residual -2, gradient [-4, -4]. New parameters [1.4, 1.4] give **prediction** 2.8, residual -1.2 and loss 1.44, down from 4.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension	Evidence required
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Apply a negative gradient

Starting with parameter a = 2, gradient -3 and **learning rate** 0.2, calculate the new a.

Teacher explanation: New a = 2 - 0.2 times -3 = 2.6.

W2 Detect overshooting

For the lesson start [1, 0], apply rate 0.3 and calculate **prediction** and loss.

Teacher explanation: Gradient [-4, -2] gives [2.2, 0.6], **prediction** 5 and loss 4, larger than the starting loss 1.

W3 Preserve one snapshot

Explain why recomputing the bias gradient after changing w is not the **simultaneous update** specified here.

Teacher explanation: The two gradient entries then belong to different parameter vectors. Save the full old gradient and use it for both assignments.

W5 Evaluate a zero gradient

At a point with gradient [0, 0], what does this update do? Does that alone prove a global minimum on any loss surface?

Teacher explanation: It leaves parameters unchanged for a finite **learning rate**. Zero gradient alone does not establish a global minimum; stationary points can have other meanings.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: [1.2, 0.1] gives **prediction** 2.5, residual -0.5 and loss 0.25. Exit: no.

The mixed-state procedure lowers this loss but does not implement the specified simultaneous gradient step; compare states and gradients, not just whether loss decreased.

W1 Apply a negative gradient

W2 Detect overshooting

W3 Preserve one snapshot

W4 Update a new model independently

Use x = 1, target 4, w = 1, b = 1, squared loss and **learning rate** 0.1. Compute **prediction**, residual, gradient [2 times residual times x, 2 times residual], new parameters and new loss.

Teacher explanation: **Prediction** 2, residual -2, gradient [-4, -4]. New parameters [1.4, 1.4] give **prediction** 2.8, residual -1.2 and loss 1.44, down from 4.

W5 Evaluate a zero gradient

Independent assessment and fresh retry

W4 requires gradient [-4, -4], parameters [1.4, 1.4] and loss 1.44. Fresh retry: x = 1, target 1, w = 1, b = 1, rate 0.25 gives residual 1, gradient [2, 2], new parameters [0.5, 0.5] and loss 0.

Require both updates to use the original residual.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect,support and revisit the mathematics

> Use the saved evidence to choose the next step.

Project and next-week bridge

Add a labelled updating parameters **evidence** sheet to Image-pattern Network Investigation. Show how you can apply a **learning rate** to an ordered gradient.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Use separate old, gradient and new columns. Extension: for the lesson example, show that the new residual is the old residual multiplied by (1 - 10 times rate).

Explain why rates 0 and 0.2 preserve loss while 0.3 raises it for a nonzero residual.

Week 12 repeats the **training** cycle for a small network.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Delayed return and support record

Suggested return points in this level: ai-5/12, ai-5/14, ai-5/17. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Apply a **learning rate** to an ordered gradient

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: For the lesson start [1, 0], apply rate 0.3 and calculate **prediction** and loss.

Reasoned practice answer: Teacher explanation: Gradient [-4, -2] gives [2.2, 0.6], **prediction** 5 and loss 4, larger than the starting loss 1.

Check every learner before discussion: Use the worked case for Updating parameters. Proposed explanation: “Subtracting a negative gradient decreases the parameter”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Subtracting a negative adds a positive amount; show the signed calculation.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Set **learning rate** 0.2 from the same starting point. Require a **prediction** and a reason before comparing. Expected result: New w=2.6, b=1.8, y=7, squared loss 4: no improvement.

Reason: The larger update jumps to an equally distant point on the opposite side of the target.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

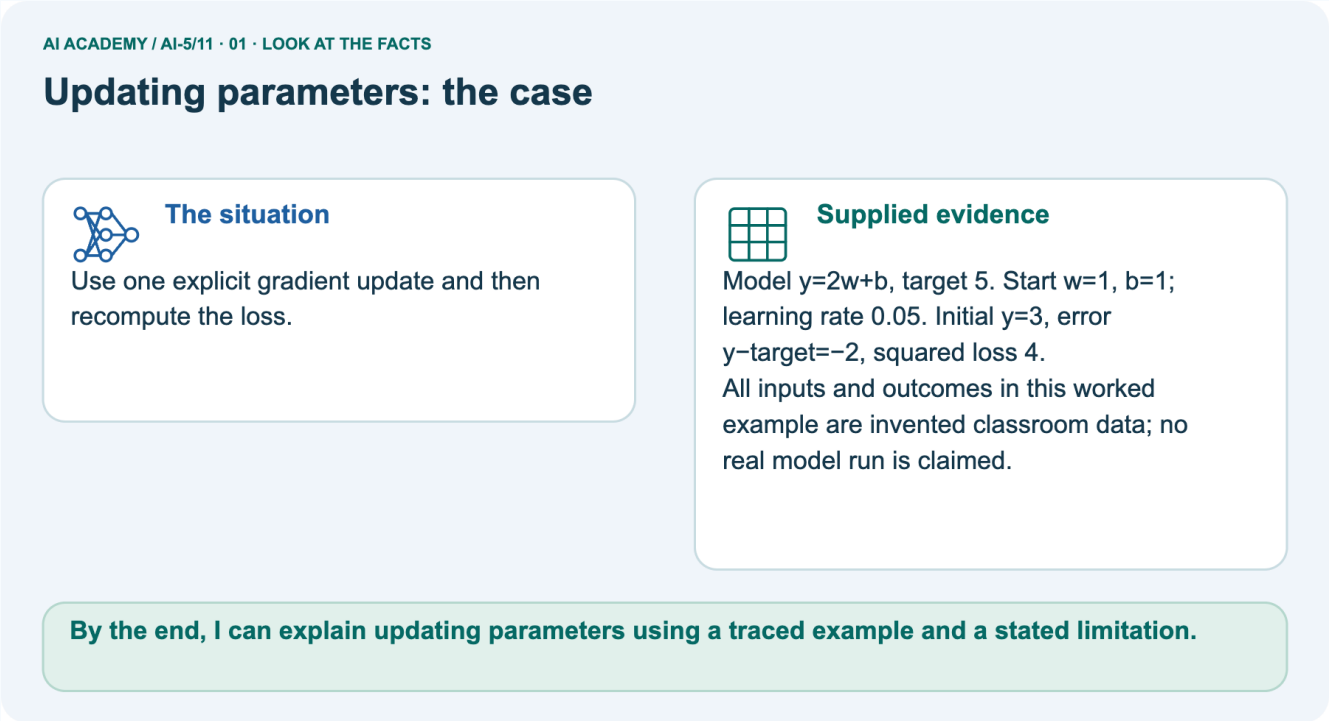
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Updating parameters: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

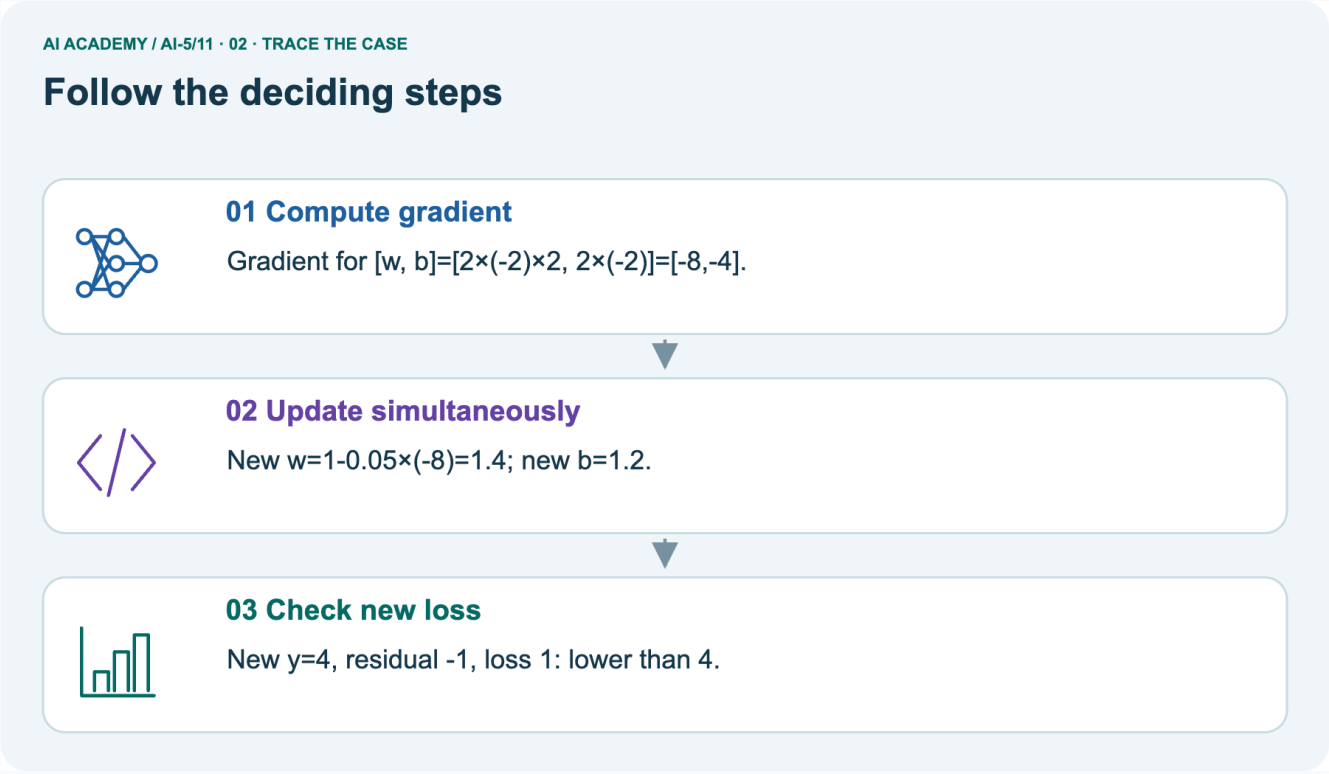
Use one explicit gradient update and then recompute the loss. Supplied facts: Model y=2w+b, target 5. Start w=1, b=1; **learning rate** 0.05. Initial y=3, error y−target=−2, squared loss 4.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



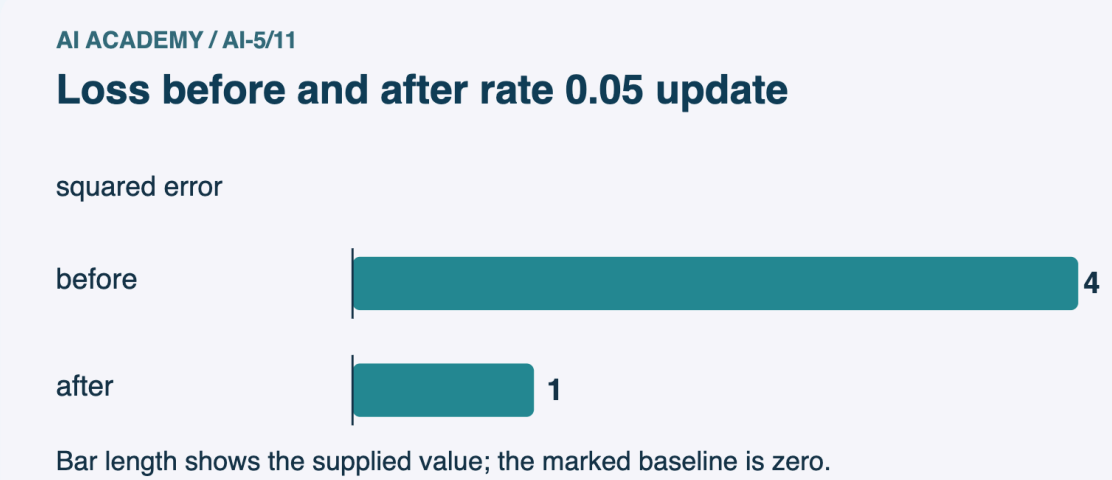
Follow the labelled picture with the matching explanation. Open the vector diagram

Compute gradient: Gradient for [w, b]=[2×(-2)×2, 2×(-2)]=[-8,-4]. → Update simultaneously: New w=1-0.05×(-8)=1.4; new b=1.2. → Check new loss: New y=4, residual -1, loss 1: lower than 4.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Loss before and after rate 0.05 update



Follow the labelled picture with the matching explanation. Open the vector diagram

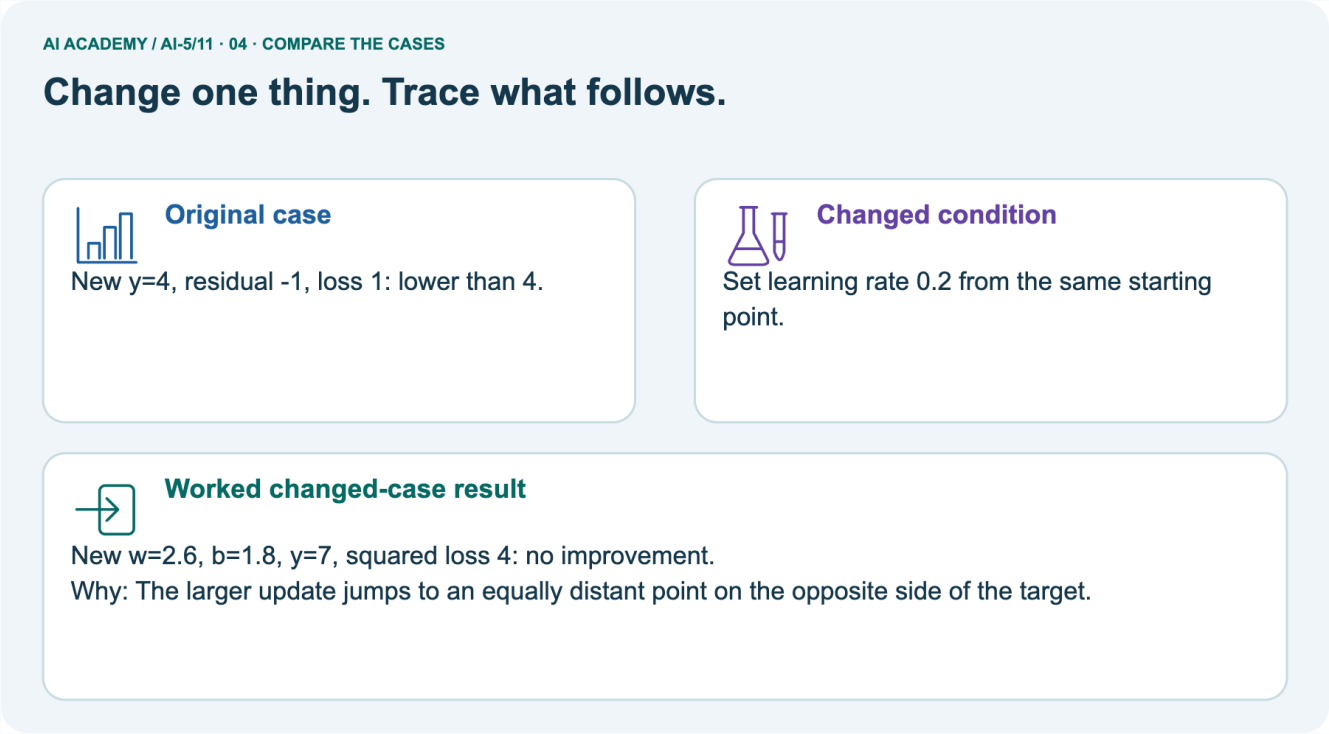
Loss before and after rate 0.05 update. Units: squared error. before: 4. after: 1.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: New y=4, residual -1, loss 1: lower than 4. Changed condition: Set **learning rate** 0.2 from the same starting point. Expected result: New w=2.6, b=1.8, y=7, squared loss 4: no improvement.

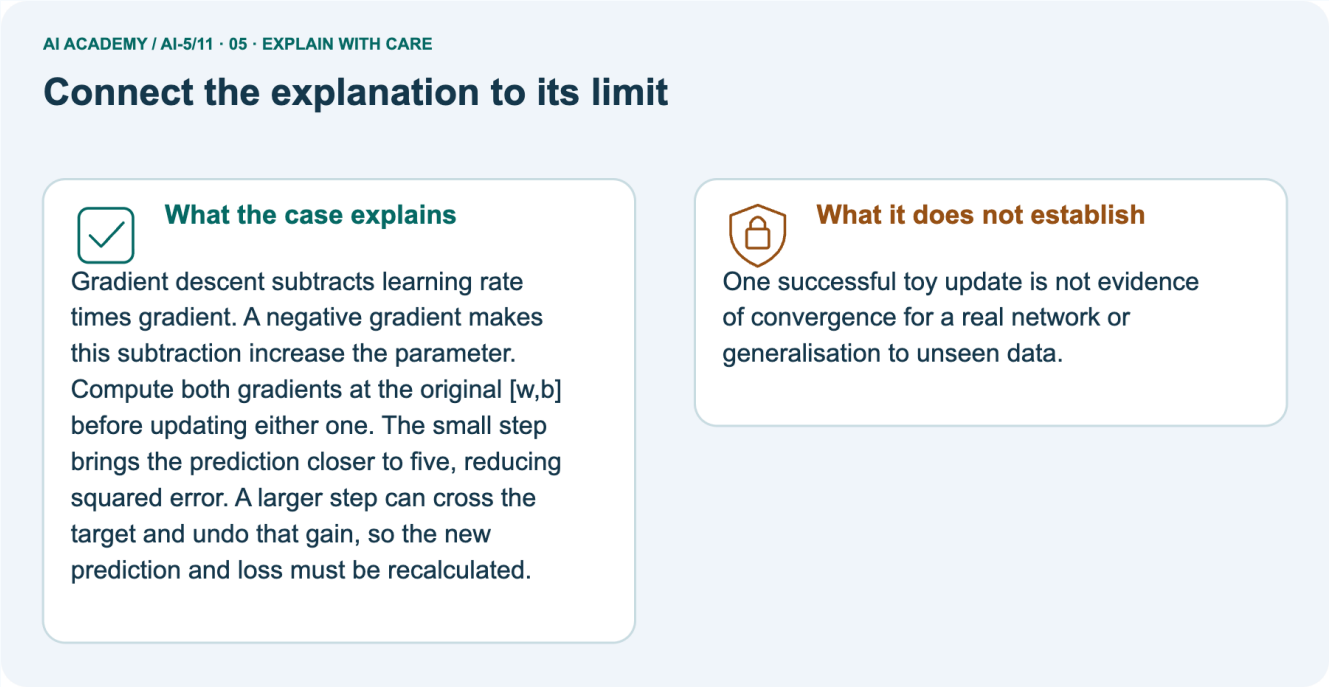
Reason: The larger update jumps to an equally distant point on the opposite side of the target.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The larger update jumps to an equally distant point on the opposite side of the target.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Gradient descent subtracts **learning rate** times gradient. A negative gradient makes this subtraction increase the parameter. Compute both gradients at the original [w,b] before updating either one.

The small step brings the **prediction** closer to five, reducing squared error. A larger step can cross the target and undo that gain, so the new **prediction** and loss must be recalculated.

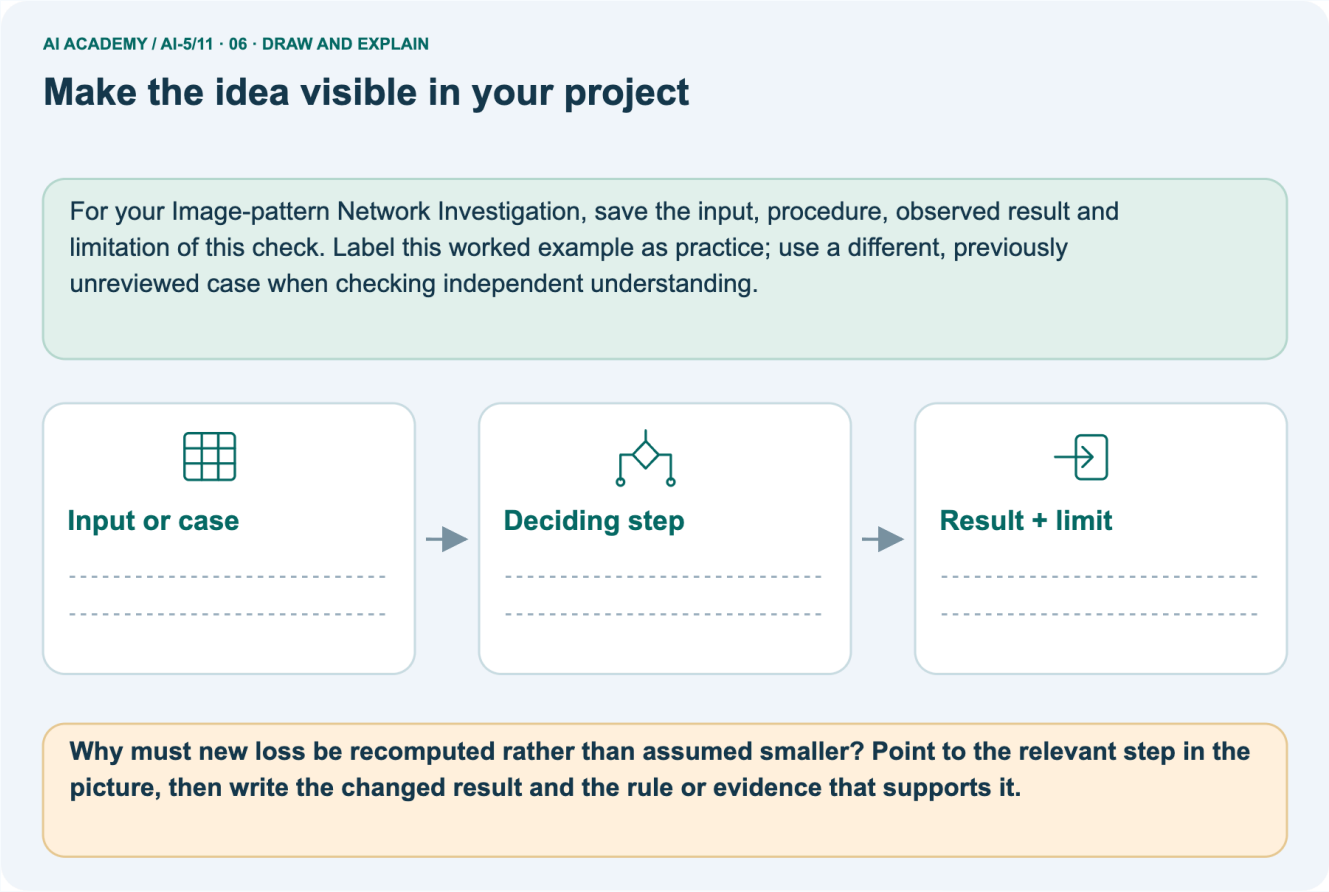
Limit: One successful toy update is not **evidence** of convergence for a real network or generalisation to unseen data.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: One successful toy update is not **evidence** of convergence for a real network or generalisation to unseen data.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why must new loss be recomputed rather than assumed smaller? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: A gradient supplies a local direction, but step size can **overshoot**. Rate 0.2 gives the same loss 4 despite using the correct gradient.

Changed-case check: New w = 2.6, b=1.8, y = 7, loss 4: no improvement. The larger update jumps to an equally distant point on the opposite side of the target.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

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▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.